NAME: SANTHOSH D

REG NO: (192412209)

COURSE NAME: ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 8a	**SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB**
DATE:	

Objective

i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.

ii) To Plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.

iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

Matlab code

```
clc;
clear;
close all;
%% Objective 1: Generate BPSK symbols
N = 1000;                       % Number of bits
data = randi([0 1], N, 1);      % Random binary sequence
% BPSK Mapping
bpsk_symbols = 2*data - 1;
%% Objective 2: Plot BPSK Signal Constellation
figure;
subplot(1,2,1);
scatter(real(bpsk_symbols), imag(bpsk_symbols), ...
```

```
    30, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-1.5 1.5]);
ylim([-1 1]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');
%% Objective 3: Add Noise and Evaluate Performance
SNR = 5;                    % Signal-to-Noise Ratio (dB)
rx_signal = awgn(bpsk_symbols, SNR, 'measured');

subplot(1,2,2);
scatter(real(rx_signal), imag(rx_signal), ...
    15, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-2.5 2.5]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);
%% Symbol Detection
```

```
detected_bits = real(rx_signal) > 0;
% Calculate Errors
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of Bit Errors       : %d\n', num_errors);
fprintf('Bit Error Rate (BER)       : %f\n', BER);
```

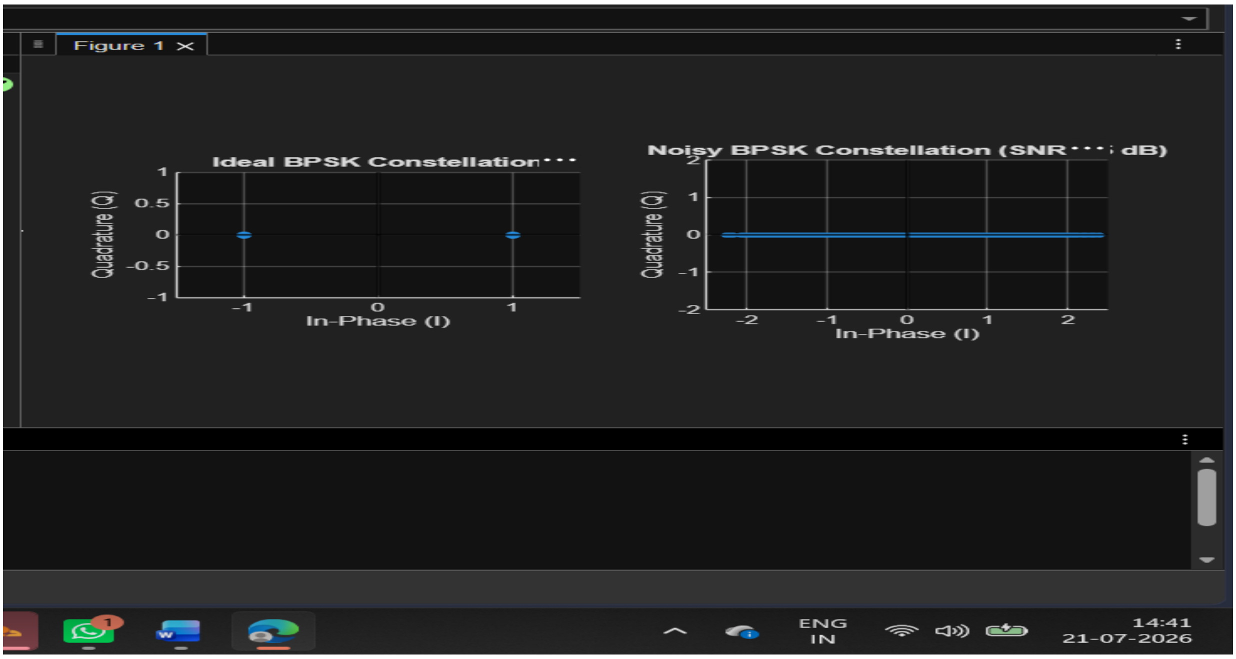
Procedure:

Objective 1: Generate BPSK Symbols from a Binary Data Sequence Using Phase-Shift Keying Principles

1. Initialize the MATLAB environment and define the number of bits.
2. Generate a random binary sequence consisting of 0s and 1s.
3. Apply BPSK modulation by mapping binary 0 to −1 and binary 1 to +1.
4. Store the generated BPSK symbols for further processing.
5. Verify that the symbol sequence contains only two distinct symbol values.
6. Prepare the modulated symbols for constellation analysis.

Objective 2: Plot and Analyze the BPSK Signal Constellation in the In-Phase (I) and Quadrature (Q) Plane

1. Use the generated BPSK symbols as inputs for constellation plotting.
2. Assign the real part of the symbols to the In-Phase (I) axis.
3. Set the Quadrature (Q) component to zero for all symbols.
4. Plot the constellation points on the I–Q plane using a scatter plot.
5. Draw the horizontal and vertical axes through the origin and label them appropriately.
6. Analyze the constellation diagram and identify the symbol locations corresponding to binary 0 and binary 1.



Result : The BPSK constellation was successfully plotted in the In-Phase (I) and Quadrature (Q) plane, showing two distinct symbol points corresponding to binary 0 and binary 1 for the given SNR.